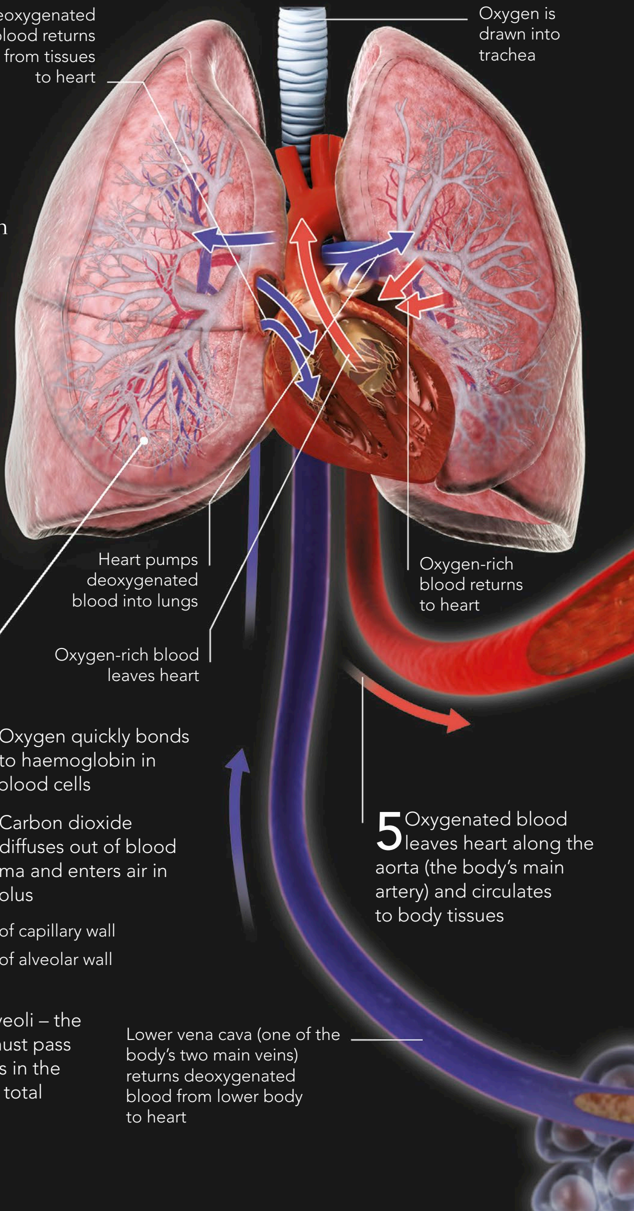
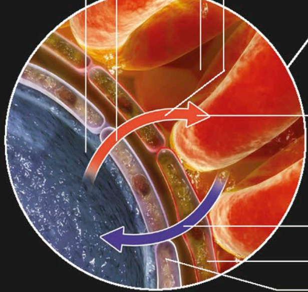
GAS EXCHANGE

THE BODY CANNOT STORE OXYGEN AND NEEDS CONTINUING SUPPLIES. IT ALSO CONSTANTLY PRODUCES CARBON DIOXIDE AS A WASTE PRODUCT. GAS EXCHANGE SWAPS OXYGEN AND CARBON DIOXIDE IN THE LUNGS AND TISSUES.

Oxygen is drawn into the body by the expanding lungs. When it reaches the ends of the lungs' airways, the gas dissolves into the fluid lining the alveoli (air sacs). It passes into the blood for distribution to each body cell. Inside cells, the oxygen reacts with glucose to free its energy. Toxic carbon dioxide is a byproduct of the process, but gas exchange discharges it into the air. In the lungs and body tissues, gases pass by diffusion: flowing from regions of high to low density.





EXCHANGE IN THE LUNGS

When fresh, oxygen-rich air reaches the alveoli – the tiny dead-end air spaces in the lungs – it must pass through several layers to reach the red cells in the blood. But these layers are so thin that the total distance is only 0.001mm ($^{1}/_{2500}$ in).